

CS F364: Design & Analysis of Algorithms

Quiz 3 — June 5, 2026

Coverage: Lectures 1–12

Note

Instructions:

- Time: 15 minutes.
- Write your answers clearly on paper, scan, and upload to Google Classroom.
- Show all steps for full credit.

Question

Consider the general recurrence

$$T(n) = T(an) + T(bn) + O(n)$$

where $0 < a, b < 1$. The deterministic selection algorithm (Median of Medians) yields such recurrences; for example, with groups of 5 we get $T(n) \leq T(n/5) + T(7n/10) + O(n)$, where $a + b = 9/10 < 1$.

(a) [2.5 marks] Let $a + b < 1$. Solve the recurrence using the substitution method (guess and verify). What is the asymptotic growth of $T(n)$?

(b) [2.5 marks] Now let $a + b = 1$. Solve the recurrence using the substitution method.

Solution

Part (a) — $a + b < 1$

Construct the recursion tree:

- **Root level:** work = cn for some constant c .
- **Level 1:** two subproblems of size an and bn . Total work = $c(an + bn) = c(a + b)n$.
- **Level 2:** four subproblems of sizes a^2n, abn, ban, b^2n . Total work = $c(a^2 + 2ab + b^2)n = c(a + b)^2n$.

At level k : total work = $c(a + b)^k n$.

Since $a + b < 1$, the work forms a **geometric series** that converges:

$$\text{Total} = cn \sum_{k=0}^{\infty} (a + b)^k = \frac{cn}{1 - (a + b)} = O(n)$$

Thus $T(n) = O(n)$.

Part (b) — $a + b = 1$

The recursion tree structure is the same, but now $a + b = 1$.

- **Root level:** work = cn .
- **Level 1:** work = $c(a + b)n = cn$.
- **Level 2:** work = $c(a + b)^2n = cn$.
- **Level k :** work = $c(a + b)^kn = cn$.

Every level contributes cn work. The tree depth is determined by the smaller branch:

$$\text{Depth} = O(\log_{1/\max(a,b)} n) = O(\log n)$$

Total work:

$$\text{Total} = cn \cdot O(\log n) = O(n \log n)$$

Key Difference

Case	Work per level	Series	Growth
$a + b < 1$	Geometric decay	Converges	$O(n)$
$a + b = 1$	Constant (cn)	Linear in depth	$O(n \log n)$

In the context of Median of Medians: groups of 5 give $a + b = 9/10 < 1$ ($O(n)$), while groups of 3 would give $a + b = 2/3 + 1/3 = 1$ ($O(n \log n)$), which is why groups of 5 (or 7) are necessary for linear-time selection.